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Ex 3: Water Jug Problem

CODE:

```
from collections import deque


def water_jug(jug1_cap, jug2_cap, target):
  visited = set()
  q = deque([((0, 0), [(0, 0)])])

  while q:
    (j1, j2), path = q.popleft()
    if j1 == target or j2 == target:
      return path

    if (j1, j2) in visited:
      continue
    visited.add((j1, j2))

    moves = [
      (jug1_cap, j2), (j1, jug2_cap),
      (0, j2), (j1, 0),
      (j1 - min(j1, jug2_cap - j2), j2 + min(j1, jug2_cap - j2)),
      (j1 + min(j2, jug1_cap - j1), j2 - min(j2, jug1_cap - j1))
```

```
    ]
    for m in moves:
      if m not in visited:
        q.append((m, path + [m]))

path = water_jug(4, 3, 2)
for step in path:
  print(step)
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-3.py ===
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
```